

Inventory Tracking Report

Project Title

Inventory Tracking Dashboard using Power BI

Project Overview

The Inventory Tracking Dashboard is designed to monitor and analyze inventory performance across products, categories, and warehouses. The dashboard provides a centralized view of stock availability, inventory value, low-stock alerts, and warehouse-level inventory distribution. It helps businesses optimize inventory management, reduce stock shortages, and improve operational efficiency.

Objectives

The primary objectives of this project are:

- Monitor current inventory levels across all products.
 - Track inventory value and stock movement.
 - Identify products with low inventory levels.
 - Compare inventory performance across warehouses.
 - Analyze inventory distribution by category.
 - Monitor inventory trends over time.
 - Provide actionable insights for inventory planning and replenishment.
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Dataset Information

Dataset: Zepto Inventory Dataset

The dataset contains inventory-related information such as:

- Product Name
 - Product Category
 - Warehouse Location
 - Quantity Available
 - Unit Price
 - Stock Date
 - Supplier Information
 - Product Status
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Data Preparation

Data Cleaning

The following data-cleaning steps were performed:

- Removed duplicate records.
- Handled missing values.
- Standardized category names.
- Corrected inconsistent warehouse entries.
- Converted date columns into proper date format.
- Verified quantity and pricing values.

Data Transformation

- Created inventory value calculation.
- Categorized products into stock status groups.
- Created date hierarchy for trend analysis.
- Generated low inventory indicators.

Key Performance Indicators (KPIs)

Total Stock Quantity

Measures the total inventory available.

DAX Measure

```
Total Stock =  
SUM(Inventory[Quantity])
```

Total Inventory Value

Measures the monetary value of inventory.

DAX Measure

```
Inventory Value =  
SUMX(  
    Inventory,
```

```
        Inventory[Quantity] * Inventory[Unit Price]
    )
```

Average Unit Price

DAX Measure

```
Average Unit Price =
AVERAGE(Inventory[Unit Price])
```

Low Inventory Count

Counts products below the defined threshold.

DAX Measure

```
Low Inventory Count =
COUNTROWS(
    FILTER(
        Inventory,
        Inventory[Quantity] <= 20
    )
)
```

Stock Turnover

Measures inventory movement efficiency.

Formula

```
Stock Turnover =
Cost of Goods Sold ÷ Average Inventory
```

Higher turnover indicates efficient inventory utilization.

Low Inventory Alert System

A low inventory flag was created to identify products requiring replenishment.

DAX Column

```
Low Inventory Flag =  
IF(  
    Inventory[Quantity] <= 20,  
    "Low Stock",  
    "Healthy Stock"  
)
```

Alert Levels

Quantity Range	Status
0 - 20	Low Stock
21 - 50	Moderate Stock
Above 50	Healthy Stock

Dashboard Visualizations

1. KPI Cards

Displays:

- Total Stock
- Inventory Value
- Average Unit Price
- Low Inventory Count

Purpose: Quick overview of inventory health.

2. Stock by Category

Visualization: Clustered Bar Chart

Displays inventory quantity for each category.

Insights:

- Highest stocked categories
 - Lowest stocked categories
 - Inventory distribution
-

3. Warehouse Comparison

Visualization: Column Chart

Compares inventory quantities across warehouses.

Insights:

- Warehouse performance
 - Inventory concentration
 - Stock allocation effectiveness
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4. Inventory Value by Category

Visualization: Treemap

Shows inventory value contribution by category.

Insights:

- High-value inventory groups
 - Revenue-generating categories
-

5. Inventory Trend Over Time

Visualization: Line Chart

Tracks inventory movement over time.

Insights:

- Seasonal trends
 - Inventory growth patterns
 - Stock fluctuations
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6. Low Inventory Alert Visual

Visualization: Table with Conditional Formatting

Highlights products with low stock.

Indicators:

- Red → Low Stock
 - Yellow → Moderate Stock
 - Green → Healthy Stock
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7. Detailed Inventory Table

Displays:

- Product Name
- Category
- Warehouse
- Quantity
- Unit Price
- Inventory Value
- Stock Status

This visual provides detailed inventory-level analysis.

Dashboard Filters (Slicers)

The following slicers were implemented:

Category Slicer

Allows users to filter by product category.

Warehouse Slicer

Filters inventory by warehouse location.

Product Slicer

Enables product-specific analysis.

Date Slicer

Supports trend analysis over selected periods.

Business Insights

The dashboard provides valuable inventory insights such as:

Inventory Availability

- Identify products with sufficient stock.
- Detect stock shortages early.

Category Performance

- Understand category-wise inventory distribution.
- Optimize category-level procurement.

Warehouse Efficiency

- Compare inventory across locations.
- Improve warehouse stock balancing.

Financial Impact

- Track total inventory value.
- Monitor high-value inventory holdings.

Replenishment Planning

- Prioritize products nearing stock depletion.
 - Reduce stock-out risks.
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Benefits of the Dashboard

Operational Benefits

- Real-time inventory monitoring
- Improved stock visibility
- Faster decision-making

Financial Benefits

- Reduced carrying costs
- Better inventory utilization
- Lower stock-out losses

Strategic Benefits

- Improved forecasting
 - Better warehouse management
 - Data-driven inventory planning
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Dashboard Layout

Top Section

KPI Cards:

- Total Stock
- Inventory Value
- Average Unit Price
- Low Inventory Count

Middle Section

- Stock by Category Chart
- Warehouse Comparison Chart

Bottom Section

- Inventory Trend Line Chart
- Low Inventory Alert Table
- Detailed Inventory Report

Right Panel

Slicers:

- Category
 - Warehouse
 - Product
 - Date
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Conclusion

The Inventory Tracking Dashboard provides a comprehensive solution for monitoring inventory performance across products, categories, and warehouses. By combining KPI tracking, low-stock alerts, trend analysis, warehouse comparisons, and inventory valuation, the dashboard enables businesses to make informed decisions regarding stock management and replenishment planning.

The solution improves inventory visibility, reduces operational risks, and supports efficient inventory control through interactive Power BI reporting.